

Taylor-Informed Indirect Adaptive Predictive Control Using Jacobian-Frozen LPV–ARX Surrogates

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Abstract—This paper presents a Taylor-informed indirect adaptive predictive control framework for nonlinear sampled-data systems based on Jacobian-frozen linear-parameter-varying autoregressive with exogenous input (LPV–ARX) surrogates. The sampled nonlinear dynamics are approximated by a finite Taylor expansion whose polynomial coefficients are identified online using recursive least squares (RLS). At each sampling instant, the Jacobian of the identified map is evaluated at the current operating point and frozen over the prediction horizon to construct an LPV–ARX predictor for model predictive control. Unlike generic nonlinear feature dictionaries, the proposed polynomial dictionary is derived directly from the Taylor structure of the sampled dynamics, enabling structure-informed pruning of monomials known a priori to vanish and thereby reducing the number of identified parameters. Numerical simulations on an unstable nonlinear benchmark compare different Taylor degrees and show that higher-order models improve tracking accuracy as the operating point moves farther from the expansion point while maintaining comparable control effort. The complete MATLAB implementation is publicly available to facilitate reproducibility.

I. INTRODUCTION

Indirect adaptive predictive control combining online identification with model predictive control (MPC) enables online adaptation from streaming input–output data while retaining the computational structure of linear predictive control [1], [2], [3]. At each sampling instant, the predictive model is updated recursively and used to compute the future control sequence over a finite horizon. Recent developments have extended this framework to a growing range of aerospace and robotic applications [4], [5], [6].

Extending adaptive predictive control to nonlinear systems remains an active research topic. Existing approaches include nonlinear model predictive control, lifting techniques based on observable dictionaries, and adaptive predictive methods employing nonlinear regressors [7], [8], [9], [10]. More recently, nonlinear extensions of the predictive cost adaptive control (PCAC) framework [11] have been investigated using kernel, polynomial, Fourier, and spline feature dictionaries [12], [13]. One approach is to replace the linear regressor by a richer feature dictionary and continue estimating the corresponding coefficients through recursive least squares (RLS). Polynomial, kernel, and other nonlinear observable representations have therefore been investigated as a means of increasing model expressiveness while preserving linearity in the parameters [14], [15], [16]. Although these dictionaries

often improve approximation capability, they are frequently selected from general families of basis functions whose connection to the underlying nonlinear plant is indirect.

The present work adopts a different perspective. Rather than introducing a generic nonlinear dictionary, we construct the feature set directly from the Taylor expansion of the sampled-data dynamics. The resulting monomials have a clear analytical interpretation, corresponding to the local polynomial structure of the nonlinear flow map. Furthermore, when the plant possesses known structural properties, monomials that are known a priori to vanish can be eliminated before identification. The resulting structured dictionary therefore reduces the number of estimated parameters while preserving the terms that characterize the local nonlinear dynamics. In this sense, the proposed dictionary is not merely expressive but also structure-informed.

After identifying the polynomial coefficients online using RLS, the identified sampled nonlinear map is transformed into a locally affine predictive model by evaluating its Jacobian at the current operating point and freezing the resulting LPV–ARX coefficients over the prediction horizon. This construction avoids repeated nonlinear propagation while retaining a predictor that adapts online through the identified Taylor coefficients. The resulting surrogate is then used within a conventional MPC formulation.

The objective of this paper is to investigate, through numerical simulation, the influence of the Taylor degree on identification and closed-loop tracking performance. In particular, we compare structured Taylor dictionaries of different orders and evaluate the trade-off between model complexity and control performance on a nonlinear benchmark system. To facilitate reproducibility, the complete MATLAB implementation used throughout this study is publicly available at <https://github.com/tamwng/taylor-informed-iapc>.

II. SYSTEM DESCRIPTION

Consider the continuous-time nonlinear system

$$\dot{x} = f(x, u), \quad (1)$$

where $f : \mathbb{R}^n \times \mathbb{R}^m \rightarrow \mathbb{R}^n$, $x \in \mathbb{R}^n$ is the state, and $u \in \mathbb{R}^m$ is the control.

Under zero-order hold with sampling period $T_s > 0$,

$$u(t) = u_k, \quad t \in [kT_s, (k+1)T_s),$$

where $t \in \mathbb{R}_{\geq 0}$ is continuous time and $k \in \mathbb{N}_0$ is the sampling index. The exact sampled-data dynamics are

$$x_{k+1} = F(x_k, u_k), \quad (2)$$

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where $F : \mathbb{R}^n \times \mathbb{R}^m \rightarrow \mathbb{R}^n$ denotes the sampled-data flow map associated with (1).

Let (x_e, u_e) denote a selected equilibrium pair satisfying

$$f(x_e, u_e) = 0. \quad (3)$$

Then,

$$F(x_e, u_e) = x_e.$$

Define the shifted variables

$$\tilde{x} := x - x_e, \quad \tilde{u} := u - u_e.$$

The shifted sampled-data dynamics are

$$\tilde{x}_{k+1} = \bar{F}(\tilde{x}_k, \tilde{u}_k), \quad (4)$$

where

$$\bar{F}(\tilde{x}, \tilde{u}) := F(x_e + \tilde{x}, u_e + \tilde{u}) - x_e. \quad (5)$$

By construction,

$$\bar{F}(0, 0) = 0. \quad (6)$$

The following assumptions are used throughout this paper.

Assumption 1. The state x_k is available for feedback at each time k .

Assumption 2. $\bar{F}$ is well-defined on a neighborhood $\mathcal{N} \subset \mathbb{R}^{n+m}$ of the origin.

Assumption 3. $\bar{F} : \mathcal{N} \rightarrow \mathbb{R}^n$ is of class $\mathcal{C}^{D+1}$ on $\mathcal{N}$.

III. TAYLOR APPROXIMATION OF THE SAMPLED DYNAMICS

Under Assumption 3, $\bar{F} : \mathcal{N} \rightarrow \mathbb{R}^n$ admits a finite Taylor approximation around the origin.

Let

$$p := n + m, \quad z := \begin{bmatrix} \tilde{x} \\ \tilde{u} \end{bmatrix} \in \mathbb{R}^p. \quad (7)$$

Since $\bar{F}(0) = 0$, the order- D Taylor approximation is

$$\bar{F}(z) = Mz + \sum_{d=2}^D F_d(z) + r_D(z), \quad (8)$$

where

$$M := \left. \frac{\partial \bar{F}}{\partial z} \right|_{z=0} \in \mathbb{R}^{n \times p}, \quad (9)$$

and $r_D : \mathcal{N} \rightarrow \mathbb{R}^n$ is the Taylor remainder. For each $d \in \{2, \dots, D\}$, $F_d : \mathbb{R}^p \rightarrow \mathbb{R}^n$ collects all homogeneous polynomial terms of total degree d . Thus, F_2 contains all quadratic terms, F_3 contains all cubic terms, and so forth. For $D = 1$, the summation in (8) is empty.

Equation (8) shows that the local Taylor approximation is a polynomial in the components of z . Define the full Taylor index set

$$\mathcal{A}_D := \{\alpha \in \mathbb{N}_0^p : 1 \leq |\alpha| \leq D\}, \quad |\alpha| := \sum_{j=1}^p \alpha_j. \quad (10)$$

For each $\alpha \in \mathcal{A}_D$, define the monomial

$$z^\alpha := \prod_{j=1}^p z_j^{\alpha_j}, \quad (11)$$

where z_j is the j -th component of z .

Let $\mathcal{A}_D = \{\alpha_1, \dots, \alpha_q\}$. The polynomial regressor is

$$\phi_D(z) := [z^{\alpha_1} \quad \dots \quad z^{\alpha_q}]^\top \in \mathbb{R}^q. \quad (12)$$

The regressor ϕ_D contains all pure and crossed monomials of total degree from 1 to D . Therefore,

$$q = \binom{p+D}{D} - 1. \quad (13)$$

Accordingly, the order- D Taylor model is written compactly as

$$\bar{F}(z) = \Theta \phi_D(z) + r_D(z), \quad (14)$$

where $\Theta \in \mathbb{R}^{n \times q}$ contains the Taylor coefficients.

IV. RECURSIVE TAYLOR COEFFICIENT IDENTIFICATION

The Taylor model (14) is linear in the coefficient matrix Θ . Neglecting the Taylor remainder, identifying the sampled dynamics reduces to estimating the coefficient matrix Θ .

For each $k \geq 0$, define

$$z_k := \begin{bmatrix} \tilde{x}_k \\ \tilde{u}_k \end{bmatrix}, \quad \phi_k := \phi_D(z_k). \quad (15)$$

The a priori one-step prediction of $\tilde{x}_{k+1}$ is

$$\hat{\tilde{x}}_{k+1} = \hat{\Theta}_k \phi_k. \quad (16)$$

Vectorize the coefficient matrix as

$$\theta_k := \text{vec}(\hat{\Theta}_k) \in \mathbb{R}^{nq}, \quad (17)$$

and define the regression matrix

$$\varphi_k := \phi_k^\top \otimes I_n \in \mathbb{R}^{n \times nq}. \quad (18)$$

Then (16) becomes

$$\hat{\tilde{x}}_{k+1} = \varphi_k \theta_k. \quad (19)$$

The prediction error is

$$e_k := \tilde{x}_{k+1} - \varphi_k \theta_k. \quad (20)$$

Using standard RLS with forgetting factor $\lambda \in (0, 1]$ and $P_0 \succ 0$, define

$$L_k = \lambda^{-1} P_k, \quad (21)$$

$$P_{k+1} = L_k - L_k \varphi_k^\top (I_n + \varphi_k L_k \varphi_k^\top)^{-1} \varphi_k L_k, \quad (22)$$

$$\theta_{k+1} = \theta_k + P_{k+1} \varphi_k^\top e_k. \quad (23)$$

The pair (P_{k+1}, θ_{k+1}) is the posterior RLS estimate obtained after processing the measured transition $(z_k, \tilde{x}_{k+1})$. Thus, once $\tilde{x}_{k+1}$ is measured, θ_{k+1} is available for the subsequent control synthesis.

Using this posterior estimate, reshape θ_{k+1} into $\hat{\Theta}_{k+1} \in \mathbb{R}^{n \times q}$ and define the identified sampled Taylor map

$$\hat{F}_{k+1}(z) := \hat{\Theta}_{k+1} \phi_D(z). \quad (24)$$

Thus, $\hat{F}_{k+1}$ is obtained from (14) by replacing the unknown coefficient matrix Θ with its RLS estimate $\hat{\Theta}_{k+1}$ and neglecting the Taylor remainder.

The map $\hat{F}_{k+1}$ is nonlinear in z but linear in the identified coefficients. This structure enables direct RLS identification. Other RLS variants, including variable-rate forgetting or covariance reset, can be used without changing the Jacobian-frozen construction below.

In the sequel, $\hat{F}_k$ denotes the most recent identified map available at the current control instant.

V. JACOBIAN-FROZEN LPV-ARX SURROGATE

The identified Taylor map $\hat{F}_k$ is nonlinear in the shifted state and input. Exact finite-horizon propagation of $\hat{F}_k$ would lead to nonlinear predictive control. Instead, we construct a local affine surrogate by freezing the Jacobian of $\hat{F}_k$ at the current operating point.

A. Operating Point

At step k , the shifted state $\tilde{x}_k$ is measured and $\tilde{u}_k$ is the currently applied input. The control computation produces the subsequent input $\tilde{u}_{k+1}$. Accordingly, the operating point is chosen as

$$\bar{z}_k := z_k = \begin{bmatrix} \tilde{x}_k \\ \tilde{u}_k \end{bmatrix} \in \mathbb{R}^{n+m}. \quad (25)$$

The initial input u_0 is prescribed.

The Taylor approximation remains centered at the equilibrium $z = 0$. The operating point $\bar{z}_k$ is used only for Jacobian evaluation.

B. Jacobian Compilation

Compute the Jacobian of the identified Taylor map at $\bar{z}_k$:

$$J_k := \left. \frac{\partial \hat{F}_k}{\partial z} \right|_{z=\bar{z}_k} \in \mathbb{R}^{n \times (n+m)}. \quad (26)$$

Partition

$$J_k = \begin{bmatrix} A_k & B_k \end{bmatrix}, \quad (27)$$

where $A_k \in \mathbb{R}^{n \times n}$ and $B_k \in \mathbb{R}^{n \times m}$ are the Jacobians of $\hat{F}_k$ with respect to the shifted state and input, respectively.

The first-order local approximation of $\hat{F}_k$ at $\bar{z}_k$ is

$$\hat{F}_k(z) \approx \hat{F}_k(\bar{z}_k) + J_k(z - \bar{z}_k). \quad (28)$$

Substituting $z = [\tilde{x}^\top \ \tilde{u}^\top]^\top$ and partitioning J_k according to (27) yields

$$\hat{F}_k(z) \approx c_k + A_k \tilde{x} + B_k \tilde{u}, \quad (29)$$

where

$$c_k := \hat{F}_k(\bar{z}_k) - A_k \tilde{x}_k - B_k \tilde{u}_k. \quad (30)$$

The Jacobian-frozen predictive surrogate is therefore

$$\tilde{x}_{i+1|k} = c_k + A_k \tilde{x}_{i|k} + B_k \tilde{u}_{i|k}, \quad i = 0, \dots, N-1. \quad (31)$$

The matrices A_k , B_k , and c_k are frozen over the horizon.

C. Reduction to Linear PCAC

The proposed scheme contains linear PCAC as its first-order special case.

Corollary 1: If $D = 1$, then the Jacobian-frozen surrogate (31) reduces to the linear PCAC propagation model.

Proof: For $D = 1$, the Taylor regressor satisfies $\phi_1(z) = z$. Hence,

$$\hat{F}_k(z) = \hat{\Theta}_k z, \quad (32)$$

where $\hat{\Theta}_k \in \mathbb{R}^{n \times (n+m)}$. Therefore,

$$J_k = \hat{\Theta}_k = \begin{bmatrix} A_k & B_k \end{bmatrix} \quad (33)$$

is independent of the operating point. Moreover,

$$c_k = \hat{F}_k(\bar{z}_k) - J_k \bar{z}_k \quad (34)$$

$$= \hat{\Theta}_k \bar{z}_k - \hat{\Theta}_k \bar{z}_k = 0. \quad (35)$$

Substituting $c_k = 0$ into (31) yields

$$\tilde{x}_{i+1|k} = A_k \tilde{x}_{i|k} + B_k \tilde{u}_{i|k}, \quad (36)$$

which is the linear PCAC propagation model. ■

D. Elimination of Factor-Freezing Ambiguity

The Jacobian freeze avoids arbitrary choices in mixed nonlinear terms. For example, consider

$$F(y, u) = \alpha y + \beta u + au^2 + buy + cy^2. \quad (37)$$

At $(\bar{y}, \bar{u})$, the Jacobian-frozen surrogate is

$$F(y, u) \approx C_{\text{loc}} + A_{\text{loc}} y + B_{\text{loc}} u, \quad (38)$$

where

$$A_{\text{loc}} = \alpha + b\bar{u} + 2c\bar{y}, \quad (39)$$

$$B_{\text{loc}} = \beta + 2a\bar{u} + b\bar{y}, \quad (40)$$

$$C_{\text{loc}} = F(\bar{y}, \bar{u}) - A_{\text{loc}} \bar{y} - B_{\text{loc}} \bar{u}. \quad (41)$$

Thus, the cross term satisfies

$$uy \approx \bar{u}y + \bar{y}u - \bar{u}\bar{y}. \quad (42)$$

This approximation matches the value and Jacobian at the operating point. It does not require choosing whether to freeze u or y .

VI. PREDICTIVE CONTROL SYNTHESIS

Following the PCAC formulation [11], the Jacobian-frozen surrogate (31) is used over a prediction horizon N .

At step k , initialize

$$\tilde{x}_{0|k} = \tilde{x}_k, \quad \tilde{u}_{0|k} = \tilde{u}_k. \quad (43)$$

The predicted states satisfy

$$\tilde{x}_{i+1|k} = c_k + A_k \tilde{x}_{i|k} + B_k \tilde{u}_{i|k}, \quad i = 0, \dots, N-1. \quad (44)$$

Here, $\tilde{u}_{0|k}$ is known, while $\tilde{u}_{1|k}, \dots, \tilde{u}_{N-1|k}$ are decision variables. This indexing follows the PCAC convention, where $\tilde{u}_{1|k}$ is applied at step $k+1$.

Let

$$\Delta \tilde{u}_{i|k} := \tilde{u}_{i|k} - \tilde{u}_{i-1|k}, \quad i = 1, \dots, N-1, \quad (45)$$

and let $\tilde{r}_{i|k}$ denote the reference in shifted coordinates.

Assume that the state and input constraint sets are the convex polyhedra

$$\mathcal{X} = \{\tilde{x} \in \mathbb{R}^n : H_x \tilde{x} \leq h_x\}, \quad (46)$$

$$\mathcal{U} = \{\tilde{u} \in \mathbb{R}^m : H_u \tilde{u} \leq h_u\}. \quad (47)$$

To soften the state constraints, introduce $\varepsilon_{i|k} \geq 0$ and solve

$$\begin{aligned} & \underset{\substack{\tilde{x}_{1|k}, \dots, \tilde{x}_{N|k}, \\ \tilde{u}_{1|k}, \dots, \tilde{u}_{N-1|k}, \\ \varepsilon_{1|k}, \dots, \varepsilon_{N|k}}}{\text{minimize}} & \frac{1}{2} \sum_{i=1}^N \|\tilde{x}_{i|k} - \tilde{r}_{i|k}\|_{Q_i}^2 \\ & + \frac{1}{2} \sum_{i=1}^{N-1} \|\Delta \tilde{u}_{i|k}\|_R^2 + \frac{1}{2} \sum_{i=1}^N \|\varepsilon_{i|k}\|_S^2 \quad (48) \\ & \text{subject to} \quad (44), \\ & H_x \tilde{x}_{i|k} \leq h_x + \varepsilon_{i|k}, \quad i = 1, \dots, N, \\ & H_u \tilde{u}_{i|k} \leq h_u, \quad i = 1, \dots, N-1, \\ & \varepsilon_{i|k} \geq 0, \quad i = 1, \dots, N, \end{aligned}$$

where $Q_i \succeq 0$, $R \succ 0$, $S \succ 0$, and $\|v\|_Q^2 := v^\top Q v$.

Since A_k , B_k , and c_k are fixed over the horizon, (48) is a convex quadratic program. The slack variables prevent infeasibility caused solely by the state constraints. No recursive-feasibility or closed-loop stability guarantee is asserted because the identified model and its Jacobian-frozen surrogate vary online.

Let $\tilde{u}_{1|k}^*$ denote the first optimized input. The computed control is

$$u_{k+1} = u_e + \tilde{u}_{1|k}^*. \quad (49)$$

VII. CASE STUDY: UNSTABLE TRIGONOMETRIC PLANT

A. Plant and Sampled Dynamics

Consider the continuous-time scalar system

$$\dot{x} = f(x, u) := a \sin x \cos x + b \cos^2 x \sin u, \quad (50)$$

where $a > 0$ and $b > 0$. Select $(x_e, u_e) = (0, 0)$ as the equilibrium. Hence, $\tilde{x} = x$ and $\tilde{u} = u$, and tildes are omitted throughout this section. Near the origin,

$$\dot{x} = ax + bu + \mathcal{O}(\|(x, u)\|^3), \quad (51)$$

so the equilibrium is open-loop unstable and its linearization is controllable.

Under zero-order hold with sampling period T_s , let F denote the exact sampled-data flow map. For the numerical study, we use the forward-Euler discretization

$$\begin{aligned} x_{k+1} &= F_{\text{FE}}(x_k, u_k) \\ &:= x_k + T_s (a \sin x_k \cos x_k + b \cos^2 x_k \sin u_k). \end{aligned} \quad (52)$$

Near the origin,

$$x_{k+1} = (1 + aT_s)x_k + bT_s u_k + \mathcal{O}(\|(x_k, u_k)\|^3), \quad (53)$$

whose open-loop eigenvalue is $1 + aT_s > 1$.

Next, since

$$f(-x, -u) = -f(x, u), \quad (54)$$

uniqueness of solutions implies that the exact flow map and its forward-Euler approximation satisfy

$$\begin{aligned} F(-x, -u) &= -F(x, u), \\ F_{\text{FE}}(-x, -u) &= -F_{\text{FE}}(x, u). \end{aligned}$$

Their Taylor expansions about the origin therefore contain only odd-total-degree terms, which enables structure-informed coefficient pruning.

B. Taylor Structure and Coefficient Pruning

Let

$$z = [x \ u]^\top. \quad (55)$$

Expanding (52) about the origin through degree five yields

$$\begin{aligned} F_{\text{FE}}(x, u) &= (1 + aT_s)x + bT_s u \\ &\quad - \frac{2aT_s}{3}x^3 - bT_s x^2 u - \frac{bT_s}{6}u^3 \\ &\quad + \frac{2aT_s}{15}x^5 + \frac{bT_s}{3}x^4 u + \frac{bT_s}{6}x^2 u^3 + \frac{bT_s}{120}u^5 \\ &\quad + \mathcal{O}(\|z\|^7). \end{aligned} \quad (56)$$

By (55), all even-total-degree terms vanish. Furthermore, the structure of (52) implies that, for a degree- D truncation, only monomials of the form

$$x^{2i+1}, \quad x^{2i}u^{2j+1}, \quad i, j \in \mathbb{N}_0, \quad (57)$$

with total degree at most D are retained. This yields a structured Taylor dictionary.

The resulting regressors are

$$\begin{aligned} \varphi_1^{\text{FE}} &= [x \ u]^\top, \\ \varphi_3^{\text{FE}} &= [x \ u \ x^3 \ x^2 u \ u^3]^\top, \\ \varphi_5^{\text{FE}} &= [x \ u \ x^3 \ x^2 u \ u^3 \ x^5 \ x^4 u \ x^2 u^3 \ u^5]^\top, \\ \varphi_7^{\text{FE}} &= [(\varphi_5^{\text{FE}})^\top \ x^7 \ x^6 u \ x^4 u^3 \ x^2 u^5 \ u^7]^\top. \end{aligned}$$

Table I compares the number of coefficients in the full Taylor dictionary and the structured dictionary.

For example, according to (56), the analytical coefficient vector associated with φ_5^{FE} is

$$\theta_5^* = \begin{bmatrix} 1+aT_s \\ bT_s \\ -2aT_s/3 \\ -bT_s/6 \\ 2aT_s/15 \\ bT_s/3 \\ bT_s/6 \\ bT_s/120 \end{bmatrix}. \quad (58)$$

Thus, structure-informed pruning reduces the number of estimated coefficients from 20 to 9 for $D = 5$, and from 35

TABLE I
NUMBER OF ESTIMATED TAYLOR COEFFICIENTS.

Degree D	Full dictionary	Structured dictionary
1	2	2
3	9	5
5	20	9
7	35	14

to 14 for $D = 7$. The analytical vector θ_5^* is used only as a reference in the numerical evaluation; the controller estimates the retained coefficients online using RLS.

No constant monomial is estimated; the affine term arises subsequently from Jacobian freezing.

C. Identified Map and Jacobian-Frozen Surrogate

For each Taylor degree $D \in \{1, 3, 5, 7\}$, let φ_D^{FE} denote the structured forward-Euler regressor defined by the monomial pattern above. The identified sampled map is

$$\hat{F}_{D,k}(x, u) = \hat{\theta}_{D,k}^\top \varphi_D^{\text{FE}}(x, u), \quad (59)$$

where $\hat{\theta}_{D,k}$ is updated online using RLS.

At each step k , the Jacobian-frozen coefficients are evaluated at the current operating point (x_k, u_k) as

$$A_k = \left. \frac{\partial \hat{F}_{D,k}}{\partial x} \right|_{(x_k, u_k)}, \quad (60)$$

$$B_k = \left. \frac{\partial \hat{F}_{D,k}}{\partial u} \right|_{(x_k, u_k)}, \quad (61)$$

$$c_k = \hat{F}_{D,k}(x_k, u_k) - A_k x_k - B_k u_k. \quad (62)$$

For the fifth-order structured regressor φ_5^{FE} , let $\hat{\theta}_{ij,k}$ denote the estimated coefficient multiplying $x^i u^j$. The resulting Jacobian coefficients are

$$A_k = \hat{\theta}_{10,k} + 3\hat{\theta}_{30,k}x_k^2 + 2\hat{\theta}_{21,k}x_k u_k + 5\hat{\theta}_{50,k}x_k^4 + 4\hat{\theta}_{41,k}x_k^3 u_k + 2\hat{\theta}_{23,k}x_k u_k^3, \quad (63)$$

$$B_k = \hat{\theta}_{01,k} + \hat{\theta}_{21,k}x_k^2 + 3\hat{\theta}_{03,k}u_k^2 + \hat{\theta}_{41,k}x_k^4 + 3\hat{\theta}_{23,k}x_k^2 u_k^2 + 5\hat{\theta}_{05,k}u_k^4. \quad (64)$$

The Jacobian-frozen predictor is therefore

$$x_{i+1|k} = c_k + A_k x_{i|k} + B_k u_{i|k}, \quad i = 0, \dots, N-1. \quad (65)$$

For comparison, the analytical Jacobian of the forward-Euler map is

$$A_{\text{FE}}(x, u) = 1 + T_s [a \cos(2x) - b \sin(2x) \sin u], \quad (66)$$

$$B_{\text{FE}}(x, u) = T_s b \cos^2 x \cos u. \quad (67)$$

VIII. NUMERICAL EVALUATION

This section evaluates the effect of the Taylor degree on closed-loop tracking performance. The continuous-time plant (50) is simulated under zero-order hold using fourth-order Runge–Kutta over each sampling interval. The controller uses the Taylor-informed identified map and the Jacobian-frozen predictor developed above. All cases $D \in \{1, 3, 5, 7\}$ use the same initialization data, constraints, prediction horizon, and control weights.

The complete MATLAB implementation used throughout this study is publicly available at <https://github.com/tamwng/taylor-informed-iapc>.

TABLE II
NUMERICAL CONFIGURATION.

Quantity	Value
Plant parameters	$a = 0.5, b = 1.5$
Sampling period	$T_s = 0.05$ s
Plant integration	RK4 with ZOH input
Initial state	$x_0 = 0$
Constraints	$ x_k \leq 0.75, u_k \leq 1$
Taylor degrees	$D \in \{1, 3, 5, 7\}$
Prediction horizon	$N = 16$
Forgetting factor	$\lambda = 1$
Initial RLS estimate	$\hat{\theta}_{D,0} = 0_{q_D^{\text{FE}}}$
Initial covariance	$P_0 = 10^5 I_{q_D^{\text{FE}}}$
State weights	$Q_i = 1, i < N, Q_N = 10$
Input-increment weight	$R = 5 \times 10^{-2}$
State-slack weight	$S = 10^5$
Initialization length	$N_{\text{id}} = 60$ samples
Initialization input	$u_k \in \{0, \pm 0.15, \pm 0.30, \pm 0.45, \pm 0.60\}$
Reference type	Amplitude-swept sinusoid
Segment length	$K_r = 300$ samples
Sinusoidal frequency	0.10 Hz
Amplitudes	$\{0.25, 0.55, 0.80, 0.90, 0\}$

A. Simulation Setup

The numerical configuration is summarized in Table II. During the first N_{id} samples, a fixed multilevel input sequence is applied and the RLS estimator is updated. The multiple initialization levels excite the odd monomials retained in the structured Taylor dictionary. The predictive controller is activated at $k = N_{\text{id}}$.

After the initialization phase, the reference is an amplitude-swept sinusoid. For the j th segment, the reference is defined by

$$r_k = A_j \sin(2\pi f(k - N_{\text{id}})T_s),$$

where $f = 0.10$ Hz and

$$A_j \in \{0.25, 0.55, 0.80, 0.90, 0\}.$$

Each amplitude is maintained for $K_r = 300$ samples before transitioning to the next level. The increasing amplitudes progressively excite higher-order nonlinearities while remaining within the imposed state and input constraints.

The cases are compared using state trajectories, input trajectories, tracking errors, and quantitative performance metrics computed over each amplitude segment.

B. Closed-Loop Results

Figures 1–3 summarize the closed-loop responses obtained with the proposed controller. The state trajectories are shown in Figure 1, the corresponding tracking errors in logarithmic scale are shown in Figure 2, and the mean tracking error for each reference amplitude is summarized in Figure 3. Quantitative performance metrics are reported in Table III. The root-mean-square error (RMSE) measures the overall tracking accuracy, the mean absolute error (MAE) measures the average tracking error, the maximum state violation reports the largest violation of the imposed state constraint, and the total input variation (I.V.) quantifies the accumulated variation of the control input.

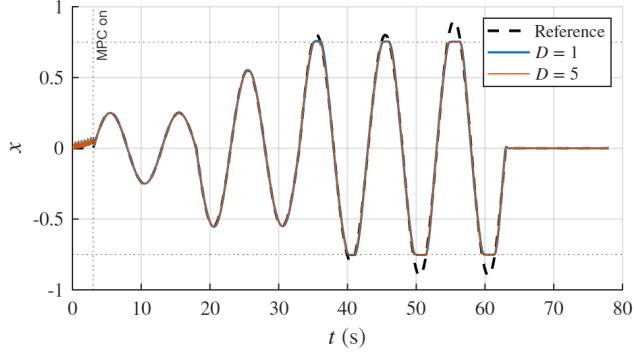


Fig. 1. Closed-loop state tracking for the first- and fifth-order Taylor models. Both controllers follow the amplitude-swept sinusoidal reference throughout the simulation. The trajectories are visually similar, while the fifth-order model remains slightly closer to the reference during the larger-amplitude segments. The horizontal dotted lines indicate the imposed state constraint.

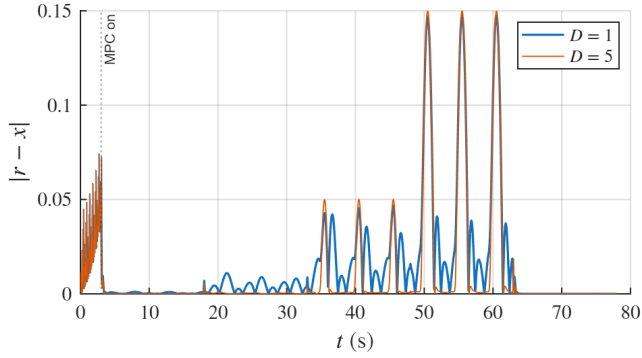


Fig. 2. Absolute tracking error in logarithmic scale for the first- and fifth-order Taylor models. The transient errors are of comparable magnitude. During each sinusoidal segment, the fifth-order model generally maintains a lower steady error level.

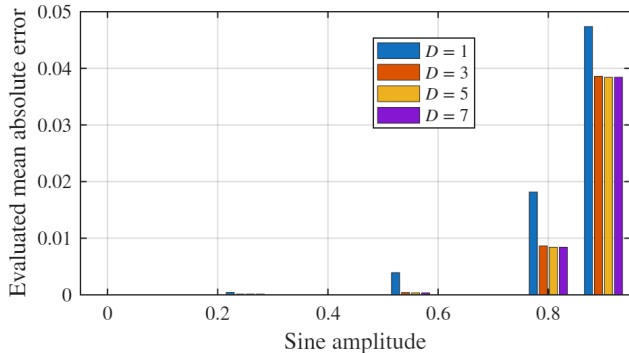


Fig. 3. Mean absolute tracking error for each reference amplitude and Taylor degree. The differences between Taylor degrees are negligible for small-amplitude motion. As the reference amplitude increases, the first-order model exhibits consistently larger mean errors than the higher-order models, whereas the third-, fifth-, and seventh-order models remain close to one another. This behavior is consistent with the increasing influence of nonlinear terms farther from the expansion point.

TABLE III
CLOSED-LOOP PERFORMANCE FOR DIFFERENT TAYLOR DEGREES D .

D	RMSE	MAE	Max. state violation	Total I.V.
1	0.03199	0.01418	7.37×10^{-3}	12.41
3	0.03092	0.00971	3.50×10^{-4}	11.75
5	0.03090	0.00958	2.56×10^{-4}	11.86
7	0.03089	0.00959	3.52×10^{-4}	11.81

Figure 1 compares the responses obtained with the first- and fifth-order Taylor models. Both controllers follow the reference throughout the simulation. However, the fifth-order model remains slightly closer to the reference during the larger-amplitude segments. The first-order model also exhibits slightly larger excursions near the imposed state constraint, whereas the fifth-order model remains marginally farther from the constraint boundary.

Figure 2 presents the corresponding absolute tracking errors in logarithmic scale. The transient errors of the two models are of similar magnitude. As the response evolves within each sinusoidal segment, the fifth-order model generally maintains a lower error level than the first-order model, indicating improved local approximation of the sampled nonlinear dynamics.

The trend is summarized quantitatively in Figure 3, which reports the mean absolute tracking error for each sinusoidal amplitude. At small amplitudes, the differences between Taylor degrees are negligible. As the amplitude increases, the first-order model exhibits consistently larger mean errors than the higher-order models, whereas the third, fifth, and seventh-order models remain close to one another. This behavior is consistent with the fact that higher-order nonlinear terms become increasingly influential as the system operates farther from the equilibrium about which the Taylor expansion is constructed.

IX. CONCLUSION

This paper presented a Taylor-informed indirect adaptive predictive control framework based on Jacobian-frozen LPV-ARX surrogates for nonlinear systems. The proposed approach first discretizes the continuous-time plant and approximates the sampled-data dynamics by a finite Taylor expansion. The resulting polynomial dictionary is identified online using recursive least squares (RLS), yielding an adaptive representation of the sampled nonlinear map. At each sampling instant, the Jacobian of the identified map is evaluated at the current operating point and frozen over the prediction horizon, producing an LPV-ARX surrogate suitable for finite-horizon model predictive control.

The numerical study demonstrated that increasing the Taylor degree improves tracking accuracy when the system operates farther from the expansion point, while maintaining comparable control effort. Furthermore, the particular structure of the sampled dynamics allows the Taylor dictionary to be pruned by eliminating monomials that are known a priori to vanish, thereby reducing the number of parameters to identify and accelerating the online RLS adaptation.

Future work will investigate variable-rate forgetting for improved adaptation under changing operating conditions, extend the proposed framework to systems with time delays [17], and validate the approach on real-world physical platforms.

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